

## GE-Problem Set 4, ECN 714

**Exercise 1** This exercise shows that the aggregate excess demand function satisfies the boundary condition. More precisely, let there be  $I$  consumers and  $L$  goods. Each consumer's preferences are strongly monotone, continuous, and strictly convex. Let  $z : \Delta^o \rightarrow \mathbb{R}^L$  be the aggregate excess demand function. Follow the steps below to prove that  $z$  satisfies

$$[p^n \in \Delta^o, p^n \rightarrow p, p \in \Delta \setminus \Delta^o] \implies \|z(p^n)\|_\infty \rightarrow \infty$$

where  $\|z\|_\infty = \max\{z_1, \dots, z_L\}$ . You may assume that  $z$  is continuous, satisfies Walras Law, and boundedness below ( $\exists x \in \mathbb{R}^L$  s.t.  $z(p) \geq x, \forall p \in \Delta^o$ ).

Proceed by contradiction, that is to say assume  $\|z(p^n)\|_\infty \nrightarrow \infty$ .

- Show that for some agent  $j$ ,  $p \cdot \omega_j > 0$ .
- For the agent  $j$ , show that  $\|z_j(p^n)\|_\infty \nrightarrow \infty$ .
- Prove that  $\exists$  a subsequence  $p^{n_k}$  s.t.  $z_j(p^{n_k}) \rightarrow x$  for some  $x \in \mathbb{R}^L$ .
- Without loss of generality, assume  $p_1 = 0$  and  $p_2 > 0$ . Prove that

$$\exists y \in \mathbb{R}^L \text{ s.t. } x + y + \omega_j \succ_j x + \omega_j \text{ and } p \cdot (x + y + \omega_j) < p \cdot (x + \omega_j)$$

- Prove that

$$\exists K \in \mathbb{N} \text{ s.t. } \forall k > K, x + y + \omega_j \succ_j z_j(p^{n_k}) + \omega_j \text{ and } p^{n_k} \cdot (x + y + \omega_j) < p^{n_k} \cdot (z_j(p^{n_k}) + \omega_j)$$

- Explain the contradiction that may be derived from the previous point. Is the proof complete?

**Exercise 2 (Gorman aggregation)** Consider an exchange economy with  $I$  agents having the same differentiable strictly increasing and strictly quasiconcave utility function but (possibly) different initial endowments. Assume that the common utility function is homogeneous of degree 1, and that the aggregate endowments of all goods are positive. You may ignore corner solutions for this problem.

- Find the Walrasian equilibria for this economy.
- Find the Pareto Optimal allocations in the 2 consumer case.
- How would your answer to (a) and (b) change if the utility function were homogeneous of degree  $k > 0$ ?

**Exercise 3** Consider an exchange economy with two goods and three agents. There are two agents of type  $a$  who have utility functions given by  $U_a = \min\{x_{1a}, x_{2a}\}$  and endowment  $\omega_a = (2, 6)$ . The type  $b$  agent has utility given by  $U_b = x_{1b} + x_{2b}$  and an endowment  $\omega_b = (6, 2)$ .

- Show that the allocation in which  $x_a^1 = (3, 3)$ ,  $x_a^2 = (5, 5)$  and  $x_b = (2, 6)$  is Pareto optimal and individually rational.
- Is the allocation described in the core? Prove your answer.
- Characterize all of the core allocations  $(x_a^1, x_a^2, x_b)$  for which  $x_b \sim_b \omega_b$ .
- Give an example of a core allocation for which  $x_b \succ_b \omega_b$  (provide *one* example and justify).
- Please find the Walrasian equilibrium price and allocation in this economy. Is this allocation in the core?

**Exercise 4** Consider a single consumer whose consumption set is the positive orthant, and preferences  $\succ$  are continuous and satisfy

$$x \gg y \implies x \succ y.$$

Suppose there is  $p \gg 0$  and a consumption bundle  $x \in \mathbb{R}_+^L$  such that

$$|p \cdot (x - \omega)| + |\inf\{p \cdot (y - \omega) : y \succ x\}| = 0.$$

Prove that  $x \in D(p, a)$ , that is to say  $x$  is in the demand set of agent  $a$  at the price  $p$ .